

Encoding $\llbracket \cdot \rrbracket : \text{CCS-VP} \rightarrow \text{CCS}$

$$\llbracket a(x).P \rrbracket = \sum_{n \in \mathbb{N}} a_n. \llbracket P\{^n/x\} \rrbracket$$

$$\llbracket \bar{a}(e).P \rrbracket = \bar{a}_m. \llbracket P \rrbracket \quad e \downarrow m$$

$$\llbracket \tau.P \rrbracket = \tau. \llbracket P \rrbracket$$

$$\llbracket P \mid Q \rrbracket = \llbracket P \rrbracket \mid \llbracket Q \rrbracket$$

$$\llbracket \sum_{i \in I} P_i \rrbracket = \sum_{i \in I} \llbracket P_i \rrbracket$$

$$\llbracket P \setminus L \rrbracket = \llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$$

$$\llbracket P[f] \rrbracket = \llbracket P \rrbracket[f'] \quad f'(a_m) = f(a)_m$$

$$\llbracket \text{if } b \text{ then } P \rrbracket = \begin{cases} \llbracket P \rrbracket & b \text{ true} \\ 0 & \text{otherwise} \end{cases}$$

$$\llbracket K(e_1, \dots, e_h) \rrbracket = K_{n_1, \dots, n_h} \quad e_i \downarrow n_i$$

$$K_{n_1, \dots, n_h} \stackrel{\text{def}}{=} \llbracket P\{^{n_1}/x_1, \dots, ^{n_h}/x_h\} \rrbracket$$

$$\text{where } K(x_1, \dots, x_h) \stackrel{\text{def}}{=} P$$

CCS Operational Semantics

$$\begin{array}{c}
\frac{}{\alpha.P \xrightarrow{\alpha} P} \text{ACT} \\
\\
\frac{P_j \xrightarrow{\alpha} P'_j}{\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j} \text{SUM} \quad j \in I \\
\\
\frac{P \xrightarrow{\alpha} P'}{P \mid Q \xrightarrow{\alpha} P' \mid Q} \text{PAR-L} \\
\\
\frac{Q \xrightarrow{\alpha} Q'}{P \mid Q \xrightarrow{\alpha} P \mid Q'} \text{PAR-R} \\
\\
\frac{P \xrightarrow{\alpha} P' \quad Q \xrightarrow{\bar{\alpha}} Q'}{P \mid Q \xrightarrow{\tau} P' \mid Q'} \text{COM} \\
\\
\frac{P \xrightarrow{\alpha} P'}{P \setminus L \xrightarrow{\alpha} P' \setminus L} \text{RES} \quad \alpha, \bar{\alpha} \notin L \\
\\
\frac{P \xrightarrow{\alpha} P'}{P[f] \xrightarrow{f(\alpha)} P'[f]} \text{REN} \\
\\
\frac{P \xrightarrow{\alpha} P'}{K \xrightarrow{\alpha} P'} \text{CONST} \quad K \stackrel{\text{def}}{=} P
\end{array}$$

For all P in CCS-VP

$$\begin{aligned} \text{(i)} \quad & P \xrightarrow{\alpha} P' \implies \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket \\ \text{(ii)} \quad & \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q \implies \exists P'. P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q \end{aligned}$$

$$\text{where } \hat{\alpha} = \begin{cases} a_m & \alpha = a(m) \\ \bar{a}_m & \alpha = \bar{a}(m) \\ \tau & \alpha = \tau \end{cases}$$

Both implications are established by *rule induction*, or induction on the depth of the derivation tree, which is the natural choice here because the transition relation is itself defined inductively by inference rules. The natural alternative would be *structural induction on P* , case-splitting on its syntactic form rather than on the last rule fired. For most forms this coincides with rule induction, but it fails cleanly at **CONST**: a constant K is a syntactic atom with no sub-term, so structural induction provides no smaller process to apply the IH to. Rule induction avoids this by measuring derivation depth, which strictly decreases at every premise, including the unfolding step inside **CONST**.

For (i) we proceed by induction on the derivation of $P \xrightarrow{\alpha} P'$, with a case analysis on the last rule applied. The *base cases* are the three axiom forms of CCS-VP (input, output, and τ prefix): for each, the encoding $\llbracket P \rrbracket$ is a concrete CCS term that admits the required transition directly via the Act or Sum rule. For the *inductive cases* (Sum, Par-L, Par-R, Com, Res, Ren, Const, Cond), the induction hypothesis supplies transitions for the shorter sub-derivation(s), and the matching CCS rule then closes the goal.

For (ii) we instead proceed by induction on the derivation of $\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q$, with a case analysis on the syntactic form of P . For each form of P , the structure of $\llbracket P \rrbracket$ uniquely determines which CCS rule could have fired last; the induction hypothesis then recovers a CCS-VP witness P' with $P \xrightarrow{\alpha} P'$, and the encoding equations confirm $\llbracket P' \rrbracket = Q$.

Throughout both directions the key invariant is that $\hat{\alpha}$ correctly bridges CCS-VP labels and CCS channels $a(m) \leftrightarrow a_m$, $\bar{a}(m) \leftrightarrow \bar{a}_m$, $\tau \leftrightarrow \tau$ ensuring that complementary actions in CCS-VP correspond to complementary channels in CCS, so that synchronization is preserved by the encoding.

$$P \xrightarrow{\alpha} P' \implies \llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket$$

Base cases

Input $a(x).P \xrightarrow{a(m)} P\{^m/x\}$:

$$\frac{\frac{a_m.\llbracket P\{^m/x\} \rrbracket \xrightarrow{a_m} \llbracket P\{^m/x\} \rrbracket}{\sum_{n \in \mathbb{N}} a_n.\llbracket P\{^n/x\} \rrbracket \xrightarrow{a_m} \llbracket P\{^m/x\} \rrbracket} \text{Sum}}{\llbracket a(x).P \rrbracket \xrightarrow{a_m} \llbracket P\{^m/x\} \rrbracket} =$$

Output $\bar{a}(e).P \xrightarrow{\bar{a}(m)} P, e \downarrow m$:

$$\frac{\frac{}{\bar{a}_m.\llbracket P \rrbracket \xrightarrow{\bar{a}_m} \llbracket P \rrbracket} \text{Act}}{\llbracket \bar{a}(e).P \rrbracket \xrightarrow{\bar{a}_m} \llbracket P \rrbracket} =$$

Silent $\tau.P \xrightarrow{\tau} P$:

$$\frac{\frac{}{\tau.\llbracket P \rrbracket \xrightarrow{\tau} \llbracket P \rrbracket} \text{Act}}{\llbracket \tau.P \rrbracket \xrightarrow{\tau} \llbracket P \rrbracket} =$$

Inductive cases

$$\frac{P_i \xrightarrow{\alpha} P'_i}{\llbracket P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_i \rrbracket} \text{IH}$$

Sum $\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j \quad j \in I$:

$$\frac{\frac{\frac{P_j \xrightarrow{\alpha} P'_j}{\llbracket P_j \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} \text{IH}}{\sum_{i \in I} \llbracket P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} \text{Sum}}{\llbracket \sum_{i \in I} P_i \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_j \rrbracket} =$$

Par-L: $P_1 \mid P_2 \xrightarrow{\alpha} P'_1 \mid P_2$

$$\frac{\frac{\frac{P_1 \xrightarrow{\alpha} P'_1}{\llbracket P_1 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \rrbracket} \text{IH}}{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \rrbracket \mid \llbracket P_2 \rrbracket} \text{Par-L}}{\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_1 \mid P_2 \rrbracket} =$$

Par-R: $P_1 \mid P_2 \xrightarrow{\alpha} P_1 \mid P'_2$

$$\frac{\frac{P_2 \xrightarrow{\alpha} P'_2}{\llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P'_2 \rrbracket} \text{IH}}{\frac{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P_1 \rrbracket \mid \llbracket P'_2 \rrbracket}{\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P_1 \mid P'_2 \rrbracket} \text{Par-R}} =$$

Com $P_1 \mid P_2 \xrightarrow{\tau} P'_1 \mid P'_2$:

$$\frac{\frac{P_1 \xrightarrow{a(m)} P'_1}{\llbracket P_1 \rrbracket \xrightarrow{a_m} \llbracket P'_1 \rrbracket} \text{IH} \quad \frac{P_2 \xrightarrow{\bar{a}(m)} P'_2}{\llbracket P_2 \rrbracket \xrightarrow{\bar{a}_m} \llbracket P'_2 \rrbracket} \text{IH}}{\frac{\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket \xrightarrow{\tau} \llbracket P'_1 \rrbracket \mid \llbracket P'_2 \rrbracket}{\llbracket P_1 \mid P_2 \rrbracket \xrightarrow{\tau} \llbracket P'_1 \mid P'_2 \rrbracket} \text{Com}} =$$

Res $P \setminus L \xrightarrow{\alpha} P' \setminus L, \alpha \notin L \cup \bar{L}$:

$$\frac{\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH}}{\frac{\llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\} \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}}{\llbracket P \setminus L \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \setminus L \rrbracket} \text{Res}} =$$

Ren $P[f] \xrightarrow{f(\alpha)} P'[f]$, where $f'(a_m) = f(a)_m$ and $\widehat{f(\alpha)} = f'(\hat{\alpha})$:

$$\frac{\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH}}{\frac{\llbracket P \rrbracket[f'] \xrightarrow{f'(\hat{\alpha})} \llbracket P' \rrbracket[f']}{\llbracket P[f] \rrbracket \xrightarrow{\widehat{f(\alpha)}} \llbracket P'[f] \rrbracket} \text{Ren}} =$$

Const $K(e_1, \dots, e_h) \xrightarrow{\alpha} R, e_i \downarrow n_i$:

$$\frac{\frac{P\{^{n_1}/_{x_1}, \dots\} \xrightarrow{\alpha} R}{K_{n_1, \dots} \xrightarrow{\hat{\alpha}} \llbracket R \rrbracket} \text{IH}}{\llbracket K(e_1, \dots, e_h) \rrbracket \xrightarrow{\hat{\alpha}} \llbracket R \rrbracket} =$$

Cond if b then $P \xrightarrow{\alpha} P', b$ true:

$$\frac{\frac{P \xrightarrow{\alpha} P'}{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} \text{IH}}{\llbracket \text{if } b \text{ then } P \rrbracket \xrightarrow{\hat{\alpha}} \llbracket P' \rrbracket} =$$

$$\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q \implies \exists P'. P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q$$

Base cases

Input $[a(x).P]$

$$\frac{}{a(x).P \xrightarrow{a(m)} P\{^m/x\}} \text{Act}$$

$$\llbracket a(x).P \rrbracket = \sum_{n \in \mathbb{N}} a_n. \llbracket P\{^n/x\} \rrbracket.$$

The only possible transition fires branch a_m for some $m \in \mathbb{N}$, so $Q = \llbracket P\{^m/x\} \rrbracket$.

Taking $P' = P\{^m/x\}$ witnesses $P \xrightarrow{a(m)} P'$ and $\llbracket P' \rrbracket = Q$.

Output $[\bar{a}(e).P]$

$$\frac{e \downarrow m}{\bar{a}(e).P \xrightarrow{\bar{a}(m)} P} \text{Act}$$

$$\llbracket \bar{a}(e).P \rrbracket = \bar{a}_m. \llbracket P \rrbracket \text{ since } e \downarrow m.$$

The unique transition via Act forces $Q = \llbracket P \rrbracket$.

Taking $P' = P$ witnesses $P \xrightarrow{\bar{a}(m)} P'$ and $\llbracket P' \rrbracket = Q$.

Silent $[\tau.P]$

$$\frac{}{\tau.P \xrightarrow{\tau} P} \text{Act}$$

$$\llbracket \tau.P \rrbracket = \tau. \llbracket P \rrbracket.$$

The unique transition via Act forces $Q = \llbracket P \rrbracket$.

Taking $P' = P$ witnesses $P \xrightarrow{\tau} P'$ and $\llbracket P' \rrbracket = Q$.

Inductive cases

$$\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q} \text{IH}$$

$$\text{Sum} \left[\sum_{i \in I} P_i \right]$$

$$\frac{\frac{\llbracket P_j \rrbracket \xrightarrow{\hat{\alpha}} Q}{P_j \xrightarrow{\alpha} P'_j \wedge \llbracket P'_j \rrbracket = Q} \text{IH}}{\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j} \text{Sum}$$

Since $\llbracket \sum_{i \in I} P_i \rrbracket = \sum_{i \in I} \llbracket P_i \rrbracket$, any transition must originate from some branch $j \in I$.

The IH recovers $P_j \xrightarrow{\alpha} P'_j$ with $\llbracket P'_j \rrbracket = Q$.

Taking $P' = P'_j$ witnesses $P \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = Q$.

Par $[P_1 \mid P_2]$

$\llbracket P_1 \mid P_2 \rrbracket = \llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket$.

Any transition of $\llbracket P_1 \rrbracket \mid \llbracket P_2 \rrbracket$ must have arisen from one of three CCS rules.

- PAR-L

$$\frac{\frac{\llbracket P_1 \rrbracket \xrightarrow{\hat{\alpha}} Q_1}{P_1 \xrightarrow{\alpha} P'_1 \wedge \llbracket P'_1 \rrbracket = Q_1} \text{IH}}{P_1 \mid P_2 \xrightarrow{\alpha} P'_1 \mid P_2} \text{Par-L}$$

The transition on the left component gives $Q = Q_1 \mid \llbracket P_2 \rrbracket$.

The IH recovers $P_1 \xrightarrow{\alpha} P'_1$ with $\llbracket P'_1 \rrbracket = Q_1$.

Taking $P' = P'_1 \mid P_2$ witnesses $P \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = \llbracket P'_1 \rrbracket \mid \llbracket P_2 \rrbracket = Q_1 \mid \llbracket P_2 \rrbracket = Q$.

- PAR-R

$$\frac{\frac{\llbracket P_2 \rrbracket \xrightarrow{\hat{\alpha}} Q_2}{P_2 \xrightarrow{\alpha} P'_2 \wedge \llbracket P'_2 \rrbracket = Q_2} \text{IH}}{P_1 \mid P_2 \xrightarrow{\alpha} P_1 \mid P'_2} \text{Par-R}$$

The transition on the right component gives $Q = \llbracket P_1 \rrbracket \mid Q_2$.

The IH recovers $P_2 \xrightarrow{\alpha} P'_2$ with $\llbracket P'_2 \rrbracket = Q_2$.

Taking $P' = P_1 \mid P'_2$ witnesses $P \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = \llbracket P_1 \rrbracket \mid \llbracket P'_2 \rrbracket = \llbracket P_1 \rrbracket \mid Q_2 = Q$.

- COM

$$\frac{\frac{\llbracket P_1 \rrbracket \xrightarrow{a_m} Q_1}{P_1 \xrightarrow{a(m)} P'_1 \wedge \llbracket P'_1 \rrbracket = Q_1} \text{IH} \quad \frac{\llbracket P_2 \rrbracket \xrightarrow{\bar{a}_m} Q_2}{P_2 \xrightarrow{\bar{a}(m)} P'_2 \wedge \llbracket P'_2 \rrbracket = Q_2} \text{IH}}{P_1 \mid P_2 \xrightarrow{\tau} P'_1 \mid P'_2} \text{Com}$$

Complementary transitions a_m and $\bar{a}_m$ map back to $a(m)$ and $\bar{a}(m)$.

The IH on each side recovers $P_1 \xrightarrow{a(m)} P'_1$ with $\llbracket P'_1 \rrbracket = Q_1$ and $P_2 \xrightarrow{\bar{a}(m)} P'_2$ with $\llbracket P'_2 \rrbracket = Q_2$.

Taking $P' = P'_1 \mid P'_2$ witnesses $P \xrightarrow{\tau} P'$ and $\llbracket P' \rrbracket = Q_1 \mid Q_2 = Q$.

Res $[P \setminus L]$

$$\frac{\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q_0} \text{IH}}{P \setminus L \xrightarrow{\alpha} P' \setminus L} \text{Res } (\alpha \notin L \cup \bar{L})$$

$\llbracket P \setminus L \rrbracket = \llbracket P \rrbracket \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$, so the transition decomposes as $\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0$ with $Q = Q_0 \setminus \{a_m \mid a \in L, m \in \mathbb{N}\}$.

The transition being unblocked forces $\hat{\alpha} \notin \{a_m \mid a \in L\}$, hence $\alpha \notin L \cup \bar{L}$.

The IH recovers $P \xrightarrow{\alpha} P'$ with $\llbracket P' \rrbracket = Q_0$.

Taking $P' \setminus L$ as the witness gives $P \setminus L \xrightarrow{\alpha} P' \setminus L$ and $\llbracket P' \setminus L \rrbracket = \llbracket P' \rrbracket \setminus \{a_m \mid a \in L\} = Q_0 \setminus \{a_m \mid a \in L\} = Q$.

Ren $[P[f]]$

$$\frac{\frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q_0} \text{IH}}{P[f] \xrightarrow{f(\alpha)} P'[f]} \text{Ren}$$

$\llbracket P[f] \rrbracket = \llbracket P \rrbracket[f']$, so the transition decomposes as $\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q_0$ with $Q = Q_0[f']$.

The IH recovers $P \xrightarrow{\alpha} P'$ with $\llbracket P' \rrbracket = Q_0$.

Taking $P'[f]$ as the witness gives $P[f] \xrightarrow{f(\alpha)} P'[f]$ and $\llbracket P'[f] \rrbracket = \llbracket P' \rrbracket[f'] = Q_0[f'] = Q$.

Cond [if b then P]

$$\frac{b \text{ true} \quad \frac{\llbracket P \rrbracket \xrightarrow{\hat{\alpha}} Q}{P \xrightarrow{\alpha} P' \wedge \llbracket P' \rrbracket = Q} \text{IH}}{\text{if } b \text{ then } P \xrightarrow{\alpha} P'} \text{Cond}$$

$\llbracket \text{if } b \text{ then } P \rrbracket = \llbracket P \rrbracket$ when b is true.

The transition is directly on $\llbracket P \rrbracket$.

The IH recovers $P \xrightarrow{\alpha} P'$ with $\llbracket P' \rrbracket = Q$.

The IH witness P' directly satisfies $P \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = Q$.

Const [$K(e_1, \dots, e_h)$]

$$\frac{e_1 \downarrow n_1, \dots, e_h \downarrow n_h \quad \frac{\llbracket P\{^{n_1}/_{x_1}, \dots\} \rrbracket \xrightarrow{\hat{\alpha}} Q}{P\{^{n_1}/_{x_1}, \dots\} \xrightarrow{\alpha} R \wedge \llbracket R \rrbracket = Q} \text{IH}}{K(e_1, \dots, e_h) \xrightarrow{\alpha} R} \text{Const}$$

$\llbracket K(e_1, \dots, e_h) \rrbracket = K_{n_1, \dots} \stackrel{\text{def}}{=} \llbracket P\{^{n_i}/_{x_i}\} \rrbracket$.

The transition is on that encoding.

The IH recovers $P\{^{n_1}/_{x_1}, \dots\} \xrightarrow{\alpha} R$ with $\llbracket R \rrbracket = Q$.

Taking $P' = R$ witnesses $K(e_1, \dots, e_h) \xrightarrow{\alpha} P'$ and $\llbracket P' \rrbracket = Q$.

Remark (implementation). The interpreter developed for this project encodes input actions over a bounded interval $[n_{\min}, n_{\max}] \subset \mathbb{N}$ rather than an infinite sum over all of $\mathbb{N}$. Concretely, $\llbracket a(x).P \rrbracket = \sum_{n=n_{\min}}^{n_{\max}} a_n \cdot \llbracket P\{^n/x\} \rrbracket$ instead of $\sum_{n \in \mathbb{N}} a_n \cdot \llbracket P\{^n/x\} \rrbracket$. The proof above remains structurally identical: the **Input** base case still applies the Sum rule and the $=$ step, and every inductive case goes through unchanged. The only difference is that operational correspondence holds under the assumption that all communicated values fall within the chosen interval; values outside it have no corresponding branch in the encoding and therefore no matching CCS transition.